# OUMA RIA SMART CLOCK

## Chapter 1 – More Than Just a Clock

*"Sometimes the simplest ideas make the biggest difference."*

---

## 1.1 How It All Began

This project did not begin as an electronics experiment.

It did not begin with an ESP32 development board, a Wi-Fi connection, or a computer program.

It began at home.

Like many families, we found ourselves relying on scraps of paper, kitchen timers, calendars and memory to keep track of everyday life. Doctor's appointments had to be remembered. Birthdays could easily be forgotten. Food cooking outside on the braai or in the smoker needed regular attention. Most importantly, medication had to be taken at the correct time every day.

As the years passed, it became clear that memory could no longer be relied upon for everything.

My wife, Ria, occasionally forgot to take her medication. We realised that a reminder needed to be reliable, visible and impossible to overlook. That simple need became the starting point for what would eventually grow into the **OUMA RIA SMART CLOCK**.

At the time, I had no idea that a simple reminder would develop into a complete home information, reminder and safety system.

---

## 1.2 Built One Problem at a Time

Unlike many electronic projects that begin with detailed specifications and design documents, this system grew naturally.

Every feature exists because a real situation demanded it.

When birthdays were forgotten, a birthday reminder was added.

When appointments became difficult to keep track of, a calendar was added.

When cooking outdoors, braai timers and smoker reminders were added.

When security became important, alarm functions were included.

When temporary antibiotic treatment required strict timing, a dedicated short-course medication scheduler was developed rather than forcing it into the normal daily medication schedule.

The project never tried to become complicated.

It simply solved one practical problem after another.

Looking back now, each small improvement became another building block until the system evolved into something much larger than originally imagined.

---

## 1.3 Designed for People, Not Technology

Many modern electronic products contain impressive technology, yet often overlook the needs of the people using them.

This project was designed with a different philosophy.

Large displays were chosen because ageing eyes appreciate clear information.

Proper tactile push buttons were selected because they can be felt without looking directly at them.

Automatic brightness adjustment allows comfortable viewing during the day and at night.

Audible reminders ensure important events are not missed.

Remote notifications allow family members to stay informed even when they are many kilometres away.

Technology should quietly assist people.

It should never become another thing they have to remember.

---

## 1.4 Life in South Africa

The OUMA RIA SMART CLOCK reflects everyday life in South Africa.

Like many homes, our family enjoys spending time around a braai. Rather than searching through recipe books or internet pages, cooking times, temperatures and useful hints are available directly from the companion web interface.

The smoker timer reminds the user to add fresh smoking wood at regular intervals during long cooking sessions.

The calendar keeps track of birthdays, anniversaries and community events.

These features may seem unusual in an electronic clock, but they reflect the way the system has always been developed—around real family life rather than around technology.

Someone in another country may replace the Braai Guide with barbecue recipes, garden reminders or local traditions. The design is flexible enough to adapt to different households and different cultures.

---

## 1.5 Built to Keep Working

South Africans understand that electrical power cannot always be taken for granted.

Frequent power interruptions influenced many design decisions throughout this project.

The system uses a real-time clock to maintain accurate time.

A backup battery allows operation to continue during power failures.

When power is restored, the clock immediately resumes normal operation without requiring the user to reset the time or reconfigure the system.

A reminder system is only useful if it remains dependable when it is needed most.

Reliability has always been considered more important than adding unnecessary features.

---

## 1.6 Sharing the Project

Originally, this system was built only for our own home.

As friends and family saw it in operation, it became clear that many of its ideas could benefit others.

Older people living independently.

Families with busy schedules.

Caregivers.

People recovering from illness.

Anyone who appreciates having important information available at a glance.

For that reason, this project is being shared openly.

It is my hope that others will build upon these ideas, adapt them to their own needs, and perhaps improve the daily lives of people in their own homes and communities.

---

## About the Author

The OUMA RIA SMART CLOCK was designed and built by **Henry Vermaak**, from the small farming community of **Koster**, in the North West Province of South Africa.

Although not a professional software developer, Henry combined decades of practical engineering experience with patience, curiosity and a willingness to learn new technologies. The project evolved over several years through continuous real-world use, with each improvement inspired by genuine day-to-day needs rather than by technical ambition.

Every feature in this system has a story behind it.

# CHAPTER 2

# System Overview

## "Every feature has a purpose."

---

## 2.1 Introduction

The **OUMA RIA SMART CLOCK** is far more than a digital clock. It is an integrated home information, reminder and safety system developed to simplify everyday life by bringing together many useful functions into a single, easy-to-use device.

Rather than requiring several separate devices for reminders, calendars, alarms, timers and notifications, the Smart Clock combines these functions into one central system that is always visible and always available.

The project has been developed over several years through practical daily use, with each new feature added only after a genuine need arose. The result is a system that is practical rather than complicated, where every function serves a useful purpose.

---

# 2.2 System Objectives

The design objectives were established from the very beginning and remained unchanged throughout the project's development.

The system was designed to:

- Provide a large, easy-to-read display suitable for users of all ages.
- Assist older people with medication reminders and daily routines.
- Help families organise appointments, birthdays and important events.
- Operate reliably during South African load-shedding conditions.
- Continue accurate timekeeping after power failures.
- Provide audible, visual and remote notifications.
- Reduce dependence on Internet services for everyday operation.
- Be simple enough that anyone in the household can use it.

Unlike many commercial smart home products, the OUMA RIA SMART CLOCK was designed so that its essential functions remain available even without an Internet connection.

---

# 2.3 Functional Overview

The Smart Clock integrates several independent subsystems into a single platform.

These include:

### Time & Date

Displays accurate time maintained by a high-accuracy Real Time Clock (RTC).

---

### Temperature Display

Continuously displays ambient temperature using the RTC temperature sensor, with colour-coded indications for quick recognition.

---

### Automatic Brightness Control

A BH1750 ambient light sensor automatically adjusts display brightness throughout the day and night for comfortable viewing while reducing power consumption.

---

### Medication Management

Supports both:

- Permanent daily medication schedules.
- Temporary medication courses (for example antibiotics).

Visual indicators, audible reminders and remote notifications work together to ensure medication is not forgotten.

---

## Calendar & Reminders

Stores birthdays, anniversaries, appointments and important family events.

---

## Egg Timer

General-purpose countdown timer for everyday household activities.

---

## Braai & Smoker Assistant

Provides cooking reference information together with smoker reminders for long cooking sessions.

Developed especially for South African outdoor cooking traditions but easily adaptable for other countries and cooking styles.

---

## Alarm & Security

Provides external alarm outputs for security applications including PIR sensors and external sirens.

Future expansion has been planned for additional alarm zones using a PCF8575 I/O expander.

---

## Audio System

The DFPlayer Mini provides spoken announcements, alarm tones and reminder sounds.

---

## Web Interface

Allows configuration and management from any device connected to the local network.

No dedicated application is required.

---

### Remote Notifications

Critical events can be sent to authorised users through the Pushover notification service.

This allows family members to receive important notifications even when they are away from home.

---

# 2.4 Hardware Architecture

The system consists of the following major hardware modules:

| Module | Purpose |
|---|---|
| ESP32-WROOM | Main processor and Wi-Fi communication |
| DS3231 RTC | Accurate timekeeping and temperature sensing |
| BH1750 | Ambient light measurement |
| WS2812 LED Matrix | Main display |
| WS2812 Status LEDs | Visual reminder indicators |
| DFPlayer Mini | Audio playback |
| Speaker | Voice prompts and alarms |
| IRLZ44N MOSFET | External alarm control |
| Buck Converter | 12 V to 5 V power conversion |
| Schottky Diode | Automatic battery backup changeover |
| Push Buttons | User interface |

# 2.5 Software Architecture

The firmware has been developed using the Arduino framework running on an ESP32.

The software consists of several independent modules working together:

- Display Manager
- Time Manager
- Medication Manager
- Calendar Manager
- Alarm Manager
- Web Server
- Wi-Fi Manager
- Audio Manager
- Notification Manager
- Configuration Manager

- Brightness Controller

This modular approach allows new features to be added without affecting the operation of the existing system.

---

# 2.6 Power System

The Smart Clock is powered from a regulated 12 V DC power supply.

A high-efficiency buck converter provides the regulated 5 V supply required by the ESP32 and LED displays.

Battery backup capability has been included to allow continuous operation during power failures.

Field testing has demonstrated that the system automatically resumes normal operation following extended power interruptions without requiring the user to reset the clock.

---

# 2.7 Design Philosophy

Every subsystem within the OUMA RIA SMART CLOCK exists because it solved a real problem encountered during everyday use.

Nothing has been included simply because the technology was available.

Each improvement has been introduced only after proving its practical value within the designer's own home.

This philosophy has produced a system that is straightforward to use, reliable in operation and capable of adapting to changing family needs over time.

# CHAPTER 3

# Hardware Architecture

*"Good hardware quietly disappears. The user notices only that everything simply works."*

---

## 3.1 Introduction

The OUMA RIA SMART CLOCK was designed using readily available electronic modules combined into a reliable and practical system.

Rather than designing complex custom electronics, proven commercial modules were selected wherever possible. This approach reduced development time, simplified maintenance and allows future builders to obtain replacement parts easily.

Each hardware component was selected because it solved a specific problem within the overall system. No module has been included without a practical purpose.

The prototype has evolved through continuous real-world testing, resulting in a hardware platform that is dependable, easy to service and readily expandable.

---

# 3.2 Hardware Philosophy

One of the principal design objectives was simplicity.

Every hardware decision was guided by the following principles:

- Use proven modules wherever possible.
- Avoid unnecessary complexity.
- Keep wiring logical and easy to understand.
- Make future expansion straightforward.
- Allow easy replacement of individual modules.
- Design for reliability before appearance.
- Ensure the system continues operating during everyday use.

Although the development prototype has a hand-built appearance, every design decision reflects these principles.

The final production enclosure is intended to present the same electronics in a cleaner mechanical design without changing the underlying hardware architecture.

---

# 3.3 Main Controller

## ESP32-WROOM

The ESP32-WROOM serves as the central controller for the entire system.

It was selected because it provides an excellent balance between processing power, memory, wireless communication and cost.

The ESP32 performs the following functions:

- Controls the LED display
- Reads all sensors
- Maintains Wi-Fi communication
- Hosts the internal web server
- Controls the audio system
- Manages medication schedules
- Operates reminder functions
- Controls security functions
- Sends remote notifications
- Performs OTA firmware updates

Throughout development the ESP32 has demonstrated excellent stability while simultaneously managing many independent software tasks.

---

# 3.4 Timekeeping

## DS3231 Real-Time Clock

Accurate timekeeping is fundamental to the operation of the Smart Clock.

The DS3231 RTC provides:

- Highly accurate timekeeping
- Calendar functions
- Battery-backed operation
- Integrated temperature sensor

During extended power interruptions the RTC continues operating from its backup battery.

When mains power returns, the ESP32 immediately retrieves the correct time from the RTC, allowing the system to resume normal operation without user intervention.

Real-world testing during prolonged South African load-shedding confirmed the effectiveness of this approach.

---

# 3.5 Display System

The visual display is the centrepiece of the entire project.

Four WS2812 LED matrix modules provide a bright, colourful and easily readable display visible from across a room.

The display presents:

- Time
- Date
- Temperature
- Animations
- Reminder messages
- Alarm indications

Automatic brightness adjustment ensures comfortable viewing under all lighting conditions.

During development one LED matrix failed following repeated power interruptions.

Rather than replacing the complete display, the damaged section was rebuilt using a WS2812 LED strip wired in a serpentine configuration.

This repair remains fully functional and demonstrates the project's philosophy of practical repair and adaptation.

---

# 3.6 Ambient Light Sensor

## BH1750

The BH1750 continuously measures ambient light levels.

Its purpose is not simply to dim the display.

Instead, it ensures that the display remains comfortable to read at all times.

Bright daylight requires increased display brightness.

Night-time requires reduced brightness.

This improves:

- User comfort
- Display readability
- LED lifetime
- Power consumption

The automatic brightness system operates continuously without requiring user adjustment.

---

# 3.7 Audio System

Audio feedback forms an important part of the reminder system.

A DFPlayer Mini provides:

- Spoken announcements
- Alarm sounds
- Reminder tones
- Audio prompts

The audio system operates independently of the ESP32, reducing processor load while allowing high-quality MP3 playback.

Practical testing during development showed that certain clone DFPlayer modules require sequentially numbered audio files.

This characteristic has been incorporated into the project documentation to assist future builders.

---

# 3.8 Power System

The complete system operates from a regulated 12 V DC supply.

Power distribution consists of:

- 12 V input supply
- Schottky diode battery changeover
- Battery backup connection
- 8 A buck converter
- 5 V distribution
- Polyfuse protection
- Local decoupling capacitors

Although theoretical LED current calculations suggest considerably higher power requirements, measurements taken during normal operation indicate substantially lower current consumption due to automatic brightness control and realistic display usage.

The chosen power supply therefore provides comfortable operating margin while maintaining excellent efficiency.

---

# 3.9 Alarm Interface

External alarm devices are controlled through an IRLZ44N logic-level MOSFET, switched directly from a GPIO pin.

A 10kΩ resistor in series with the gate limits inrush current, and a second 10kΩ resistor from gate to ground holds the MOSFET firmly off during boot — without it, the gate can float high momentarily at power-on and the siren chirps before the firmware has even started. This arrangement electrically isolates the ESP32 from external loads while providing sufficient switching capability for high-power devices such as 12 V piezo alarm sirens.

The modular approach also simplifies future expansion and maintenance.

---

# 3.10 User Controls

Large tactile push buttons provide the primary user interface.

Unlike capacitive touch controls, mechanical push buttons provide positive tactile feedback that is particularly beneficial for older users.

The larger button size improves ease of operation while reducing accidental activation.

---

# 3.11 Future Expansion

The hardware platform has been designed with future growth in mind.

Planned enhancements include:

- PCF8575 I/O expansion
- Additional PIR sensors
- Additional alarm zones
- Further home automation functions

The modular hardware architecture allows these additions without significant modification to the existing design.

---

# 3.12 Summary

The hardware architecture reflects the overall philosophy of the OUMA RIA SMART CLOCK:

**Simple where possible. Reliable where necessary. Expandable for the future.**

Each module has been selected to perform a clearly defined task while working together as part of a dependable integrated system.

# CHAPTER 4

# Software Architecture

*"Good software is invisible. The user simply notices that everything works together."*

---

## 4.1 Introduction

Although the firmware is contained within a single Arduino project, it has been designed as a collection of independent software modules, each responsible for a specific function within the Smart Clock.

This modular approach makes the firmware easier to understand, maintain and expand. New features can be added with minimal impact on the rest of the system, provided each module continues to perform its own clearly defined task.

Over time, the software has grown naturally in response to practical needs identified during daily use. As a result, the firmware reflects real-world experience rather than following a rigid theoretical design.

The software continuously manages many tasks simultaneously, including timekeeping, display control, reminder scheduling, web server communication, audio playback, sensor monitoring and user interaction.

To the user, these activities appear seamless. Internally, however, they are coordinated by carefully structured program logic running on the ESP32.

---

## 4.2 Software Design Philosophy

The firmware was developed according to several guiding principles:

- Each module should perform one primary function.
- User interaction should remain simple and intuitive.
- Time-critical operations must never block the rest of the system.
- Display updates should remain smooth and responsive.
- Configuration should be accessible through the web interface whenever possible.
- Reliability takes precedence over unnecessary features.
- New functionality should integrate with the existing system without disrupting proven operation.

This philosophy has allowed the software to expand over several years while remaining stable and maintainable.

---

## 4.3 The Main Software Modules

Rather than viewing the firmware as a single program, it is helpful to think of it as a team of specialists working together.

Each module has its own responsibility while sharing information with the others.

The major software modules include:

- System Manager
- Display Manager
- Time & RTC Manager
- Brightness Controller
- Audio Manager
- Medication Manager
- Short Course Medication Manager
- Calendar & Reminder Manager
- Egg Timer Manager
- Braai & Smoker Timer Manager
- Security Manager
- Wi-Fi Manager
- Web Server
- OTA Update Manager
- Notification Manager (Pushover)
- Configuration Manager
- Button Input Manager
- Sensor Manager

Each module contributes to the overall operation of the Smart Clock without requiring the user to understand its internal workings.

# CHAPTER 5

# System Software

*"Software is where individual electronic modules become one intelligent system."*

---

## 5.1 Introduction

The hardware provides the foundation of the OUMA RIA SMART CLOCK, but it is the firmware that transforms individual electronic modules into a practical household information and reminder system.

The firmware has been developed continuously over several years, with new functions added only after they proved useful during everyday operation.

Rather than attempting to include every possible feature from the beginning, the software evolved naturally as new requirements arose. Examples include the addition of the temporary antibiotic scheduler, improvements to the automatic brightness control, and enhancements to the reminder system.

This evolutionary approach has resulted in firmware that reflects real-world use rather than theoretical design.

---

## 5.2 Program Structure

Although the firmware is contained within a single Arduino project, it is organised into logical functional sections.

The software continuously performs many independent tasks without interrupting normal operation.

These tasks include:

- Maintaining accurate time.
- Updating the display.
- Monitoring button inputs.
- Reading sensors.
- Managing medication schedules.
- Controlling reminder LEDs.
- Playing audio announcements.
- Hosting the web server.
- Processing Wi-Fi communication.
- Sending remote notifications.
- Monitoring security inputs.
- Performing OTA updates.

Each task operates as part of the overall system, creating the appearance of a single, seamless device.

---

## 5.3 Real-Time Operation

One of the primary design goals was to ensure that the Smart Clock remains responsive at all times.

The firmware avoids unnecessary delays and instead performs frequent, short-duration tasks.

This allows the clock to:

- Continue updating the display.
- Respond immediately to button presses.
- Process reminder events on time.
- Maintain network communication.
- Play audio without interrupting other functions.

To the user, these activities appear to happen simultaneously.

---

## 5.4 A Living Program

Unlike commercial products that are finalised before release, the Smart Clock firmware continues to evolve.

New features are introduced when genuine needs arise during daily use.

Examples include:

- The addition of a temporary medication scheduler for antibiotic courses.
- Improvements to automatic brightness control.
- Expansion of reminder functions.
- New cooking timers and reference information.
- Planned support for additional security zones.

This method of development ensures that every feature has a practical purpose.

---

## 5.5 Reliability Before Features

Throughout development, stability has always taken priority over adding new functionality.

Every significant software change is tested during normal household use before becoming part of the permanent firmware.

This approach has resulted in a stable system capable of operating continuously for extended periods while maintaining accurate time, reliable reminders, and dependable network communication.

# CHAPTER 6

# Display System

*"Designed to be read easily, understood instantly, and useful throughout the day."*

---

## 6.1 Introduction

The display is the primary user interface of the OUMA RIA SMART CLOCK.

Unlike conventional digital clocks that simply show the current time, this display has been designed to present useful household information in a clear and readable format.

The display combines a large LED matrix, coloured status indicators, and automatic brightness control to provide information that can be understood from across a room.

Special consideration was given to users with reduced eyesight by using large characters, high contrast, and minimal visual clutter.

---

## 6.2 Main Time Display

The four-module LED matrix forms the main display.

During normal operation it displays:

- Current time
- Temperature
- Weather information (when available)
- Reminder messages
- System notifications
- Countdown timers
- Configuration information

Large numerals were deliberately chosen so that the display remains readable from several metres away.

---

## 6.3 Automatic Brightness Control

Display brightness is automatically adjusted using a BH1750 ambient light sensor.

The firmware continuously measures the surrounding light level and adjusts the LED brightness accordingly.

This provides several benefits:

- Comfortable viewing during daylight.

- Reduced glare at night.
- Lower power consumption.
- Longer LED life.

The brightness transition is intentionally smooth, preventing sudden distracting changes when lighting conditions vary.

---

# 6.4 Day and Night Brightness Adjustment

*(Insert the Design Note we wrote earlier.)*

---

# 6.5 Status Indicator LEDs

A separate WS2812 LED strip provides coloured status indicators.

Each LED represents a dedicated system function such as:

- Medication reminders
- Alarm status
- Timers
- Notifications
- System activity

Using independent indicator LEDs allows important information to remain visible while the main display continues showing the time.

---

# 6.6 Weekly Status Display

A row of seven indicators displays the current day of the week.

**M T W T F S S**

Only the current day is illuminated.

Although simple, this feature has proven surprisingly useful during everyday use, particularly for retirees and people whose routines do not follow a strict work-week schedule.

The current day can be confirmed instantly without navigating menus or checking a calendar.

---

# 6.7 System Health Indicators

The web interface displays several continuously updated system status indicators.

These include:

- NTP Synchronisation
- Wi-Fi Connection
- Internet Availability
- DFPlayer Communication
- System Uptime

These indicators provide immediate confirmation that the Smart Clock is operating correctly and help identify communication problems before they affect reminder or notification functions.

For example, Wi-Fi may still be connected while the Internet service is unavailable. By monitoring both conditions independently, the cause of network-related problems can be identified quickly.

# CHAPTER 7

# Medication & Reminder System

*"Helping people remember what matters, when it matters."*

---

## 7.1 Introduction

The Medication and Reminder System is one of the most important features of the OUMA RIA SMART CLOCK.

Unlike many electronic reminder systems that simply generate an alarm at a predetermined time, this system was developed from everyday household experience where reminders must remain visible until the required action has actually been completed.

The design philosophy is simple:

**A reminder is only complete when the task has been done—not when the alarm stops.**

This approach has proven particularly valuable for medication schedules, where missing or delaying a dose can have serious consequences.

---

## 7.2 The Beginning

The medication reminder system was one of the first major functions developed for the Smart Clock.

Its development began after a simple everyday incident.

One day, Ria forgot to take her medication.

Rather than relying on memory alone, the idea emerged to build a system that would continue reminding the user until the medication had actually been taken.

From that practical beginning, the reminder system gradually expanded into one of the central features of the Smart Clock.

Today it manages regular medication, temporary treatments, appointments, birthdays, household reminders, cooking timers, and many other daily activities.

---

## 7.3 Chronic Medication Schedule

Many people take medication every day for long-term conditions.

The Smart Clock allows daily medication schedules ranging from:

- Once per day
- Twice per day
- Three times per day
- Four times per day

Each reminder time can be individually programmed through the built-in web interface.

The schedule repeats automatically every day without requiring further user intervention.

---

## 7.4 Temporary Medication Courses

During development, another practical need became apparent.

Occasionally, medication must only be taken for a limited number of days—for example, a course of antibiotics.

Initially, the system only supported permanent daily medication schedules.

After Ria was prescribed a four-day antibiotic course, a dedicated temporary medication scheduler was added.

The user enters:

- Medication name
- Number of doses per day
- Times for each dose
- Duration of the course

Once the final day has been completed, the temporary schedule automatically expires.

No manual reset is required.

This prevents unnecessary reminders after the treatment has finished.

Later, a second situation arose that the original single-course design could not handle.

When Ria was prescribed two different antibiotics at the same time for pneumonia, one course scheduler was no longer sufficient — both medications needed independent tracking, with different dose counts and different schedules, running at the same time.

Rather than forcing one course to serve two different medications, the system was expanded to support two independent short courses running simultaneously. Each course tracks its own:

- Medication name
- Doses per day
- Dose times
- Total number of days

The two courses run completely independently. One can finish while the other continues, and starting, stopping or completing one course has no effect on the other.

---

## 7.5 Extra Medication

Daily life is not always predictable.

Sometimes additional medication must be taken outside the normal schedule.

Examples include:

- Pain medication
- Allergy tablets
- Emergency medication
- Additional treatment prescribed for one day only

The Smart Clock provides an "Extra Medication Today" function that allows these one-time reminders to be recorded without affecting the normal medication schedule.

---

## 7.6 Visual Reminder System

Medication reminders are indicated using dedicated WS2812 status LEDs.

Different colours and flashing patterns immediately indicate the reminder status.

Typical states include:

- Waiting for scheduled time.
- Medication due.
- Medication overdue.
- Medication acknowledged.
- Medication completed.

Because the status LEDs remain visible continuously, reminders cannot easily be overlooked.

---

## 7.7 Audio Reminders

At the scheduled time, the Smart Clock can play spoken or recorded audio reminders using the DFPlayer Mini.

This provides an additional level of notification, particularly useful when the user is not looking directly at the display.

Audio reminders complement the visual indicators without replacing them.

---

## 7.8 Remote Notifications

When connected to the Internet, reminder notifications can also be forwarded to mobile devices using the Pushover notification service.

This allows reminders to be received even when the user is away from the Smart Clock.

Remote notifications are especially useful for family members who may wish to assist elderly parents or relatives.

---

## 7.9 A Reminder Should Not Disappear

One important design decision distinguishes the Smart Clock from many ordinary alarm clocks.

Stopping the audible alarm does not necessarily mean that the task has been completed.

Visual reminders remain active until the user confirms that the medication has actually been taken.

This greatly reduces the possibility of acknowledging an alarm and then forgetting to complete the task.

---

## 7.10 Designed for Real Life

The reminder system was not created in a laboratory.

Every feature was introduced because it solved a real problem encountered during everyday use.

This practical approach has produced a reminder system that is straightforward to operate while remaining flexible enough to accommodate changing medical treatments and daily routines.

---

## Author's Note

This medication system was developed from everyday experience rather than medical theory.

Its purpose is not to replace medical advice, but to help people follow the treatment prescribed by their healthcare professionals.

If it helps even one person avoid missing an important dose of medication, then it has achieved its purpose.

---

# CHAPTER 8

# Calendar, Timers & Daily Information

*"Keeping track of life's important moments—from birthdays to a Sunday braai."*

---

## 8.1 Introduction

While the OUMA RIA SMART CLOCK was originally developed as a reminder system, it soon became clear that daily life involves much more than medication schedules.

Appointments, birthdays, cooking times, household tasks, social events and family activities all compete for our attention.

Rather than requiring separate applications or notebooks, the Smart Clock combines these functions into a single integrated information system that is available at any time from either the display or the built-in web interface.

The goal is simple:

**Make everyday information available immediately, without searching for it.**

---

## 8.2 Calendar System

The integrated calendar provides an overview of important upcoming events.

Unlike a conventional calendar that simply records dates, the Smart Clock continuously calculates the time remaining until each event.

Examples include:

- Medical appointments
- Birthdays
- Anniversaries
- Community events
- Household reminders
- Personal notes

The web interface displays both a monthly calendar and a list of upcoming events, making it easy to plan ahead.

---

## 8.3 Countdown to Events

One particularly useful feature is the automatic countdown display.

Instead of only showing the event date, the system displays information such as:

- **Doctor appointment — in 3 days**
- **Birthday — in 12 days**
- **Pension Day — in 21 days**

This provides a constant reminder without requiring the user to calculate dates mentally.

---

## 8.4 Household Reminders

Not every reminder is medical.

Many daily activities simply need to happen at the right time.

Examples include:

- School activities
- Shopping reminders
- Household maintenance
- Visitors arriving
- Community meetings
- Paying monthly accounts

The reminder system is intentionally flexible so that each household can adapt it to its own routine.

---

## 8.5 Wekkers — Independent Daily Alarms

Not every reminder fits neatly into a medication schedule or a calendar event.

Sometimes what is needed is simply a second, third or fourth daily alarm — separate from the main wake-up alarm, and separate from medication.

The Wekkers feature (Afrikaans for "alarms") provides three completely independent daily alarms, each with its own:

- Time
- Days of the week
- On/off setting

Once a Wekker sounds, it repeats every twenty seconds until it is acknowledged using the same button used for birthdays and doctor appointments, or until it has repeated six times.

This was a deliberate choice. A single short beep is easy to sleep through. A Wekker that keeps gently insisting is far more likely to actually be noticed.

Typical uses include:

- A reminder to check on something at a specific time each day
- A second wake-up alarm for a different family member
- A recurring prompt that has nothing to do with medication at all

Because each Wekker has its own day-of-week setting, one alarm can ring every weekday while another rings only on weekends — all managed independently through the web interface.

## 8.6 Cooking Timers

Cooking was one of the earliest non-medical functions added to the Smart Clock.

Simple preset timers allow frequently repeated cooking tasks to be started with minimal button presses.

Examples include:

- Eggs
- Bread
- Baking
- General kitchen timers

These timers can generate both local audio reminders and remote notifications.

---

## 8.7 Braai Timer

One feature reflects a tradition familiar to many South Africans.

A braai is rarely rushed.

Whether cooking steaks, chops or boerewors, timing influences the final result.

The Braai Timer provides preset timing reminders while allowing the cook to enjoy family and friends instead of constantly watching the fire.

The objective is not to replace experience, but to serve as a helpful guide.

---

## 8.8 Smoker Timer

Long-duration smoking requires patience.

Large cuts of meat often remain in the smoker for several hours, during which fresh smoking wood may need to be added periodically.

The Smoker Timer assists by generating reminders at predetermined intervals.

For example, a smoked pork leg may require fresh wood chunks every thirty minutes over a period of approximately five hours.

Rather than relying on memory, the Smart Clock quietly keeps track of the process.

A dedicated front-panel button also provides an instant check of elapsed smoking time directly on the main display — useful when standing in the kitchen and wondering exactly how long the meat has been in the smoker, without needing to open the web interface.

---

## 8.9 Braai Reference Guide

An interesting addition to the web interface is the Braai Reference section.

Instead of searching through cookbooks or internet pages while preparing a meal, useful information is immediately available.

The guide includes:

- Cooking times
- Fire temperature guidance
- Meat thickness recommendations
- Turning intervals
- Poultry cooking advice
- Pork cooking recommendations
- Traditional South African sauces and marinades

Although included primarily for South African users, the information can be useful to anyone interested in outdoor cooking.

---

## 8.10 Recipes That Stay With the Clock

The recipe section was included for a practical reason.

Many favourite recipes are used repeatedly over the years.

Keeping them permanently stored within the Smart Clock means they are always available from a mobile phone or computer connected to the home network.

Examples currently include:

- Sticky Rib Braai Sauce
- Classic Peri-Peri Marinade
- Traditional Braai Guidelines

The system can easily be expanded as new family recipes are added.

---

## 8.11 More Than a Timer

The cooking functions were never intended simply as kitchen timers.

They were added because cooking is often a family activity.

The Smart Clock helps remove uncertainty from the timing while allowing people to spend more time together and less time worrying about the clock.

Sometimes the best memories are made while waiting around a fire.

---

## Author's Reflection

Technology often focuses on productivity.

The OUMA RIA SMART CLOCK also recognises the importance of everyday family life.

It helps people remember appointments and medication, but it also helps prepare Sunday lunch, reminds someone to add another piece of oak to the smoker, and keeps favourite recipes close at hand.

In this way, the Smart Clock becomes more than an electronic device.

It becomes part of the rhythm of the home.

# CHAPTER 9

# Web Interface & Configuration

*"Powerful enough to configure the entire system, yet simple enough for everyday use."*

---

## 9.1 Introduction

Although the OUMA RIA SMART CLOCK operates independently as a standalone device, nearly every function can also be configured through its built-in web interface.

Any computer, tablet or smartphone connected to the same local network can access the Smart Clock using an ordinary web browser.

No special software or mobile application is required.

The web interface was designed to make configuration straightforward while keeping the display unit itself simple to operate.

---

## 9.2 Why a Web Interface?

The Smart Clock was designed primarily for daily use, not for constant programming.

Using a web browser offers several advantages:

- Large, easy-to-read screens.
- Familiar controls.
- Simple editing of schedules.
- No additional software installation.
- Works with computers, tablets and smartphones.
- Accessible from anywhere on the local network.

This approach also allows future software improvements without changing the hardware.

---

## 9.3 Main Dashboard

The Home page provides a summary of the system's current condition.

Typical information includes:

- Current time.
- Date.
- Temperature.
- Current day of the week.
- System uptime.
- NTP synchronisation status.
- Wi-Fi connection status.
- Internet connection status.
- DFPlayer communication status.

This dashboard allows the user to determine, at a glance, that the Smart Clock is operating correctly.

---

## 9.4 Menu Structure

The web interface is organised into logical sections, allowing users to quickly locate the function they need.

Typical menu items include:

- Clock Settings
- Alarm Configuration
- Medication Scheduler

- Calendar
- Notes
- Egg Timer
- Smoker Timer
- Braai Reference
- Birthday Manager
- Doctor Appointments
- OTA Firmware Update
- Configuration Settings

Each section performs a specific task while maintaining a consistent appearance throughout the interface.

---

## 9.5 Medication Configuration

Medication schedules can be programmed entirely from the web interface.

The user selects:

- Number of daily doses.
- Time of each dose.
- Temporary medication courses.
- Extra medication reminders.
- Confirmation of medication taken.

All changes are stored automatically and become active immediately.

---

## 9.6 Calendar Management

Appointments, birthdays and reminders are entered using simple web forms.

The system automatically:

- Stores the event.
- Calculates countdown days.
- Displays upcoming events.
- Generates reminder notifications.

The user does not need to calculate dates manually.

---

## 9.7 Cooking Reference

One of the more distinctive sections of the web interface contains the Braai and Smoker reference pages.

Rather than searching online while preparing a meal, cooking information remains permanently available on the local Smart Clock.

Information includes:

- Cooking times.
- Fire temperature guides.
- Smoking intervals.
- Traditional sauces.
- Family recipes.
- Practical cooking tips.

Because these pages are stored locally, they remain available even when the Internet is unavailable.

---

## 9.8 Configuration Page

The Configuration page allows adjustment of important system parameters without modifying the firmware.

Examples include:

- Display brightness.
- Day and night brightness compensation.
- Audio volume.
- Wi-Fi settings.
- Network parameters.
- Notification settings.
- Alarm behaviour.
- Sensor adjustments.

This makes it possible to adapt the Smart Clock to different homes without rewriting software.

---

## 9.9 Over-The-Air (OTA) Updates

One of the most valuable maintenance features is support for OTA (Over-The-Air) firmware updates.

Instead of removing the ESP32 from the enclosure and reconnecting programming cables, updated firmware can be transferred through the local network.

Benefits include:

- Faster software upgrades.
- No need to open the enclosure.
- Reduced wear on USB connectors.
- Easier maintenance after installation.

For future versions installed in homes belonging to relatives or friends, OTA updates provide a convenient method of distributing software improvements.

---

## 9.10 Local First

A deliberate design decision was to ensure that the Smart Clock remains useful even without Internet access.

Most daily functions continue operating normally, including:

- Clock display.
- Medication reminders.
- Calendar.
- Timers.
- Audio announcements.
- Local web interface.
- Alarm system.

Only services requiring external communication, such as NTP synchronisation, Pushover notifications and online weather information, depend on an active Internet connection.

This philosophy ensures that temporary Internet outages do not prevent the Smart Clock from performing its primary purpose.

---

## 9.11 Designed for Everyday Users

Although the Smart Clock contains sophisticated electronics, the web interface was designed for ordinary household users rather than engineers.

The objective is not to expose technical details, but to allow common tasks to be completed easily and confidently.

Large buttons, clear menus and logical organisation help make the system accessible to users of all ages.

## 9.12 Live Status Mirror

One of the unique features of the Smart Clock web interface is the **Status Bar Mirror**.

This page reproduces, in real time, the colours displayed by the physical WS2812 status LEDs on the Smart Clock itself.

Whether viewing the system from a computer, tablet or smartphone, the user sees exactly the same information that appears on the clock.

This provides two important advantages:

- The system status can be checked remotely without being physically near the clock.
- Family members can quickly understand the current operating condition without memorising LED colours.

Selecting **"What does each light mean?"** opens a built-in reference guide explaining the purpose of every indicator.

This reference remains permanently available within the web interface, eliminating the need to consult the printed manual for everyday operation.

The mirror display and reference guide work together to make the Smart Clock intuitive for both regular users and first-time visitors.

---

# Author's Reflection

Computers often become complicated because they expose every internal detail to the user.

The Smart Clock follows a different philosophy.

The complexity remains inside the firmware.

The user sees only what is needed to perform the task at hand.

Whether scheduling medication, checking an appointment or starting a cooking timer, the objective is always the same:

**Simple actions leading to dependable results.**

# CHAPTER 10

# Security, Notifications & Alarm System

*"Providing reassurance when you are home, and confidence when you are away."*

---

## 10.1 Introduction

Although the OUMA RIA SMART CLOCK began as a reminder and household information system, it gradually evolved to include basic home security functions.

The objective was never to replace a professional alarm installation. Instead, the goal was to provide an additional layer of awareness and protection, particularly for older people living alone or spending extended periods at home.

By combining motion detection, audible alarms, visual indicators and remote notifications, the Smart Clock helps users remain informed about activity around their home while keeping operation simple and intuitive.

---

## 10.2 Why Add Security?

During the development of the Smart Clock, it became clear that reminders and safety naturally belong together.

For many elderly people, peace of mind is just as important as remembering medication.

A simple movement detector at a gate, passage or entrance can provide reassurance without requiring a complex commercial alarm system.

The Smart Clock therefore combines daily assistance with practical home awareness, allowing one device to perform several useful roles.

---

## 10.3 Security Zones

The Smart Clock supports multiple independent security zones using PIR (Passive Infrared) motion sensors.

Each zone can monitor a different area of the property, such as:

- Front entrance.
- Back door.
- Passage.
- Garage.

- Workshop.
- Garden gate.

Each sensor is monitored independently, allowing future software expansion without requiring changes to the overall system design.

In practice, this is often implemented as two zones with different behaviour.

An outdoor or courtyard zone can remain active around the clock, since motion there almost always means an intruder rather than a family member — the siren activates immediately, with no delay.

An indoor zone is better suited to an "away" mode: arming provides a short delay to leave the house, and any motion detected while armed provides a further short delay before the siren sounds, allowing the system to be disarmed on returning home.

This two-zone approach — always-on outside, delayed inside — provides strong protection without constantly disarming and rearming the system every time someone is home.

---

## 10.4 Planned Expansion

The current hardware has been designed with future growth in mind.

Additional security inputs can be added using an I²C I/O expander, such as the PCF8575.

This approach allows many more sensors to be connected while using only two ESP32 communication lines.

By planning for expansion from the beginning, the Smart Clock remains flexible as household requirements change.

---

## 10.5 Alarm Indication

When motion is detected, the Smart Clock immediately provides several forms of notification.

These may include:

- Status LED colour changes.
- Messages on the LED display.
- Audible announcements.
- Local alarm activation.
- Remote notification via the Internet.

Using several forms of indication reduces the chance that an important event will be overlooked.

---

## 10.6 High-Power Alarm Output

Rather than driving a siren directly from the ESP32, the Smart Clock uses an external MOSFET driver module.

This provides complete electrical isolation between the microcontroller and the high-current alarm load.

The MOSFET module allows various external warning devices to be connected, including:

- Piezo sirens.
- Horn speakers.
- Flashing warning lights.
- External relays.

The alarm output therefore remains flexible for different installation requirements.

---

## 10.7 Why a MOSFET Driver?

The ESP32 operates with 3.3-volt logic and is designed for low-current electronic signals.

High-power devices should never be connected directly to its output pins.

By using a dedicated MOSFET driver module, the Smart Clock can safely control higher voltages and currents while protecting the ESP32 from electrical stress.

This modular approach also makes future upgrades straightforward.

---

## 10.8 Audible Alarm

The prototype uses a high-power 12-volt piezo horn connected through the MOSFET driver.

Although compact, the horn produces a very loud warning signal that can be heard over a considerable distance.

This makes it suitable for attracting attention during an alarm condition or warning occupants of unusual activity.

The alarm output can also be adapted to suit quieter indoor installations if required.

## 10.9 Remote Notifications

When Internet access is available, alarm events can also generate immediate notifications using the Pushover service.

This allows users to receive alerts while away from home.

Remote notification is especially valuable for:

- Family members.
- Caregivers.
- Elderly people living independently.
- Holiday homes.
- Small workshops.

By combining local and remote alerts, important events are less likely to go unnoticed.

---

## 10.10 Designed Around Everyday Life

The security system was designed with the same philosophy as the remainder of the Smart Clock.

Its purpose is not to create unnecessary complexity, but to quietly assist the people living in the home.

For many users, simply knowing that doors, entrances and movement are being monitored provides an additional sense of confidence.

The system therefore becomes another part of the home's daily routine rather than a separate piece of equipment.

---

## 10.11 Reliability During Power Interruptions

Power interruptions are a reality in many parts of the world.

During development in South Africa, extended periods of load shedding influenced many design decisions.

The Smart Clock is designed so that, after power is restored:

- The real-time clock preserves accurate time.
- Stored schedules remain intact.

- Security settings are retained.
- Internet time is automatically re-synchronised when available.
- Normal operation resumes without user intervention.

This automatic recovery allows the system to continue protecting and assisting the household without requiring manual reconfiguration.

---

# Design Note – Built From Experience

The security features were not developed from a commercial alarm specification.

They were developed by asking a simple question:

**"What information would help someone feel safer at home?"**

The answer was not a complicated control panel.

It was clear indicators, reliable notifications, dependable recovery after power failures, and straightforward operation.

That philosophy guided every design decision.

---

# Author's Reflection

The OUMA RIA SMART CLOCK is not intended to replace professional security systems.

Instead, it complements everyday life by combining reminders, household information and practical home awareness in one device.

For older people, confidence often comes from knowing that important things are quietly being watched over.

Sometimes the greatest value of technology is not making life more complicated, but making people feel a little more secure.

# CHAPTER 11

# Power System, Reliability & Recovery

*"Designed to keep working when the lights go out."*

This chapter is unique because it explains why the clock has been so dependable during all the Eskom load shedding you've experienced.

---

## 11.1 Introduction

A reminder system is only useful if it can be trusted.

For that reason, considerable attention was given to the design of the Smart Clock's power system.

Rather than simply supplying the required voltages, the power system was designed to maximise reliability, simplify maintenance and recover automatically after power interruptions.

This chapter describes the design philosophy behind that approach.

---

## 11.2 Single 12-Volt Supply

The entire Smart Clock is powered from a single regulated 12-volt DC power supply.

Using a single supply simplifies wiring and allows all major components to share a common power source.

The 12-volt supply powers:

- The buck converter.
- Alarm output.
- External MOSFET module.
- Battery backup system.

---

## 11.3 5-Volt Distribution

A high-current DC-DC buck converter provides the regulated 5-volt supply required by:

- ESP32 development board.
- LED matrices.
- WS2812 status LEDs.
- DFPlayer Mini.
- Other peripheral modules.

Although theoretical calculations suggest that a fully illuminated WS2812 display could require a much larger power supply, practical measurements showed that the Smart Clock normally operates well within the capability of the selected converter.

Real-world testing confirmed that an 8-amp buck converter provides reliable operation for the prototype.

---

## 11.4 Battery Backup

Because of frequent power interruptions during development, provision was made for a 12-volt backup battery.

A Schottky diode isolates the battery from the power supply while allowing automatic change-over whenever mains power fails.

When power returns, the transition back to the normal supply occurs automatically.

No user intervention is required.

---

## 11.5 Automatic Recovery

One of the most satisfying moments during development came after an extended power interruption.

When electricity returned:

- The RTC had maintained accurate time.
- The ESP32 restarted normally.
- Wi-Fi reconnected automatically.
- Internet communication resumed.
- NTP verified the system clock.
- The Smart Clock continued operating exactly as before the outage.

No buttons needed to be pressed.

No time needed to be reset.

The system simply carried on.

This behaviour became one of the principal design goals of the project.

---

## 11.6 Real-World Reliability

The Smart Clock was not tested only in a laboratory.

It proved itself under real operating conditions in rural South Africa, including repeated power interruptions and extended periods without mains electricity.

These experiences influenced many design decisions and confirmed the importance of designing for automatic recovery rather than manual intervention.

---

## 11.7 Protection

Electrical protection has been incorporated wherever practical.

These include:

- Polyfuses.
- Common grounding.
- Filter capacitors.
- High-current wiring separation.
- MOSFET isolation for external loads.

These measures improve both safety and long-term reliability.

---

## 11.8 Built for Continuous Operation

Unlike experimental prototypes intended only for demonstration, the Smart Clock was designed to remain powered continuously.

Many months of everyday operation provided valuable opportunities to identify small problems, improve the firmware and increase overall reliability.

Every software improvement and hardware modification contributed to a system that became progressively more dependable through practical experience.

---

# Author's Reflection

One lesson became clear during development.

Reliability is rarely achieved by one brilliant idea.

It is achieved by solving hundreds of small problems, one at a time.

Each better cable.

Each improved connector.

Each software correction.

Each power supply improvement.

Individually they seem insignificant.

Together they create a system that people can trust every day.

---

# CHAPTER 12

# Installation & Assembly (Part 1)

*"Building a reliable system begins long before the first wire is connected."*

---

## 12.1 Introduction

The Ouma Ria Smart Clock was developed as a practical, serviceable and reliable system rather than a compact consumer product.

Throughout development, ease of construction, maintenance and future expansion were considered just as important as appearance.

This chapter describes the recommended approach for assembling the Smart Clock. Although every builder may choose a different enclosure or internal layout, the principles presented here have been proven during the construction and operation of the original prototype.

---

## 12.2 Before You Begin

Before reaching for the soldering iron, take time to gather all the required parts.

Check every module individually before installation.

It is far easier to replace or troubleshoot a faulty module while it is still on the workbench than after it has been mounted inside an enclosure.

A recommended sequence is:

1. Test the ESP32.

2. Test the LED matrices.
3. Test the BH1750 light sensor.
4. Test the RTC module.
5. Test the DFPlayer Mini.
6. Test each push-button.
7. Test the alarm output.
8. Finally assemble the complete system.

Building and testing one section at a time greatly reduces fault-finding later.

---

## 12.3 The Prototype

The original Smart Clock was never intended to become a published project.

It was built as a working development platform using readily available materials.

Sections of blank printed circuit board were cut to size and used as mounting plates for the various modules.

Veroboard was used for the push-button assembly, while double-sided PCB material provided convenient grounding and mechanical support.

As the firmware evolved, components could easily be removed, replaced or modified.

Although visually simple, this construction method proved extremely practical during development.

---

## 12.4 Designing the Final Version

The final published version is intended for installation inside a purpose-built enclosure.

The recommended enclosure consists of:

- A 3D-printed main housing.
- A black translucent acrylic (Perspex) front panel.
- A steel base plate for stability.
- Rubber feet to prevent movement.
- Rear connectors for power and external alarm wiring.

This arrangement provides a neat appearance while maintaining easy access for servicing.

---

## 12.5 Why Black Acrylic?

At first glance, a black front panel may seem an unusual choice.

However, smoked black acrylic provides several advantages.

When the display is off, the front panel appears as a clean, glossy black surface.

When illuminated, the LED matrices shine clearly through the acrylic while the individual LEDs become less visible.

The result is a softer, more professional-looking display with improved contrast and reduced glare.

The front panel effectively acts as a diffuser, giving the illuminated characters a smoother appearance.

---

# 12.6 Internal Layout

Whenever possible, the internal layout should separate the system into logical sections.

For example:

- Power supply.
- ESP32 controller.
- Display electronics.
- Audio system.
- Sensor connections.
- External alarm connections.

Keeping these sections organised simplifies both construction and future maintenance.

Neat wiring also improves airflow and reduces the chance of accidental damage during servicing.

---

# 12.7 Mounting the Expansion Board

One component deserves special attention.

The ESP32 plugs into an expansion board that distributes power and signals to the various modules.

This board provides regulated **3.3 V** for sensors and peripherals that require it, while allowing **5 V** devices such as the DFPlayer Mini to be powered separately.

Using the expansion board greatly simplifies wiring and reduces the number of direct connections to the ESP32 itself.

It also makes replacing the ESP32 straightforward should maintenance ever be required.

---

## 12.8 Wiring Practices

One lesson learned during development deserves special emphasis.

Not all Dupont jumper leads are manufactured to the same standard.

Very inexpensive jumper leads may contain only a single fine conductor inside the insulation.

Although they appear perfectly acceptable externally, repeated movement can cause the conductor to fracture internally, resulting in intermittent faults that are extremely difficult to diagnose.

During development, several days of troubleshooting were eventually traced to this problem.

For permanent installation, it is strongly recommended that:

- Good-quality Dupont connectors are used.
- Module connections are soldered using proper stranded instrument wire.
- Wiring is secured to prevent unnecessary movement.
- Connectors are clearly labelled wherever practical.

Good wiring practice contributes significantly to the long-term reliability of the finished system.

---

## 12.9 Think Like a Service Technician

When arranging components inside the enclosure, always consider future maintenance.

Ask yourself:

- Can the ESP32 be removed without dismantling everything?
- Can the SD card be accessed easily?
- Can a sensor be replaced if it fails?
- Are all connectors clearly identifiable?
- Is there sufficient room to work with ordinary tools?

A little extra planning during construction can save many hours of work later.

---

# Design Note – The Prototype Was Never the Goal

The prototype shown throughout this book was built as a working engineering platform.

It carries the marks of continuous development, software experimentation and hardware improvements accumulated over several years.

It was never intended to be a showpiece.

The purpose of the prototype was to prove ideas, solve problems and refine the firmware until the design became reliable.

The final enclosure represents the natural evolution of that work—not a replacement for it, but the next step in its journey.

---

# Author's Reflection

Every successful project begins with a prototype.

Some prototypes are elegant.

Others are untidy.

Neither matters.

What matters is whether the prototype teaches you something.

The Ouma Ria Smart Clock prototype taught hundreds of lessons.

Every additional wire, every firmware update, every improved connector and every solved fault contributed to the system described in this book.

Without the prototype, there would be no finished design.

# CHAPTER 13

# Installation & Assembly (Part 2)

*"Good workmanship is just as important as good software."*

---

## 13.1 Cable Management

One aspect of the Smart Clock that cannot be seen once the enclosure is assembled is the internal wiring.

Although hidden from view, careful cable management contributes significantly to the reliability and serviceability of the system.

Where practical:

- Keep power wiring separate from signal wiring.
- Route cables neatly along the sides of the enclosure.
- Secure wiring using cable ties or adhesive cable mounts.
- Avoid placing strain on module connectors.
- Leave sufficient cable length to allow modules to be removed for servicing.

A neat wiring layout not only improves appearance but also makes future fault-finding considerably easier.

---

## 13.2 Connector Identification

Every connector should be clearly identified.

Simple labels or colour coding can save many hours during maintenance or future upgrades.

Typical connectors include:

- Display
- RTC
- BH1750
- DFPlayer
- Temperature sensor
- PIR inputs
- Alarm output
- Power input

Future builders—and even the original constructor several years later—will appreciate the time spent labelling connections.

---

## 13.3 Power Distribution

Although the Smart Clock uses several operating voltages, the wiring philosophy remains straightforward.

- **12 VDC** powers the main system and external alarm equipment.

- **5 VDC** from the buck converter supplies the LED displays, DFPlayer Mini and other 5 V peripherals.
- **3.3 VDC**, provided by the ESP32 expansion board, powers compatible sensors and logic modules.

Keeping these voltage rails separate helps prevent wiring mistakes and simplifies troubleshooting.

---

# 13.4 Mounting the Sensors

Sensor placement influences system performance.

### BH1750 Light Sensor

The ambient light sensor should be positioned where it measures the general room lighting rather than the light emitted by the display itself.

Direct illumination from the LEDs may cause incorrect brightness adjustment.

During development, the sensor was mounted behind a small opening in the enclosure, allowing it to respond naturally to changes in room lighting.

### Temperature Sensor

The temperature sensor should be located away from heat-producing components such as voltage regulators and power converters.

This provides a more accurate indication of room temperature.

### PIR Sensors

Passive infrared sensors should be positioned to monitor the intended area without facing direct sunlight, heaters or rapidly changing heat sources, all of which may lead to false triggering.

---

# 13.5 Building in Stages

One of the most effective techniques used during development was to build the Smart Clock in small stages.

Each section was completed, tested and verified before moving to the next.

For example:

1. ESP32 and power supply.

2. LED display.
3. RTC.
4. Temperature sensor.
5. BH1750 light sensor.
6. DFPlayer Mini.
7. Push-buttons.
8. Web interface.
9. Alarm outputs.
10. External sensors.

This incremental approach greatly simplified software development and fault diagnosis.

---

## 13.6 Test Before Closing the Enclosure

Before fitting the final cover, perform a complete operational check.

Confirm that:

- The display operates correctly.
- Brightness changes with ambient light.
- Audio playback functions normally.
- Push-buttons respond correctly.
- Wi-Fi connects successfully.
- NTP synchronises.
- The RTC maintains accurate time.
- Temperature readings are reasonable.
- All reminder functions operate correctly.
- The alarm output activates correctly.

Only after every function has been verified should the enclosure be fully assembled.

---

## 13.7 A Builder's Tip

One lesson became very clear during the development of the Smart Clock.

If something does not work, resist the temptation to change several things at once.

Instead:

- Make one change.
- Test it.
- Confirm the result.
- Only then move on to the next step.

This disciplined approach prevents confusion and makes faults much easier to locate.

## Design Note – A Prototype Never Really Ends

The original Smart Clock evolved continuously.

New features were added.

Software improved.

Hardware was refined.

Even after the prototype had been operating successfully for months, practical experience continued to suggest worthwhile improvements.

In many respects, the prototype became a permanent development platform rather than a finished product.

This philosophy is reflected throughout the design.

The Smart Clock is intended to evolve as new ideas emerge and changing needs are identified.

---

## Author's Reflection

Engineering is often imagined as a single moment of inspiration.

In reality, it is usually a process of steady improvement.

The OUMA RIA SMART CLOCK was not created in one weekend.

It grew through patience, observation and countless small refinements.

Each improvement solved one real problem.

Taken together, those improvements transformed a simple prototype into a dependable household companion.

# CHAPTER 14

# Installation & Assembly (Part 3)

*"From Prototype to Everyday Companion."*

---

## 14.1 First Power-Up

The first power-up of a newly assembled Smart Clock is an exciting milestone.

Before applying power, perform one final visual inspection.

Confirm that:

- All power connections are correctly polarised.
- No loose wires are present.
- All modules are firmly connected.
- The ESP32 is correctly seated in its expansion board.
- The microSD card is installed.
- The antenna area of the ESP32 is unobstructed.

Once satisfied, apply power.

---

## 14.2 Expected Start-Up Sequence

A correctly assembled Smart Clock should complete its startup automatically.

During the first few seconds, the following events normally occur:

1. ESP32 boots.
2. Internal peripherals initialise.
3. RTC is read.
4. BH1750 light sensor initialises.
5. Wi-Fi connection begins.
6. Internet connectivity is checked.
7. NTP synchronises the clock.
8. DFPlayer Mini is detected.
9. Display becomes active.
10. Status LEDs indicate normal operation.

No user intervention should normally be required.

---

## 14.3 Checking System Status

After startup, verify the system status using both the display and the web interface.

Important indicators include:

- Current time.
- Current temperature.
- Day-of-week indicator.
- NTP synchronisation status.
- Wi-Fi connection.
- Internet availability.
- DFPlayer status.
- Uptime counter.

These provide immediate confirmation that all major subsystems are operating correctly.

---

## 14.4 The Importance of Uptime

One of the most useful diagnostic indicators is the uptime counter.

Rather than simply showing how long the Smart Clock has been running, the uptime provides confidence that the system is operating reliably.

Long uninterrupted operating periods indicate stable firmware, dependable hardware and a healthy power supply.

During development, increasing uptime became one of the simplest measures of system reliability.

---

## 14.5 Learning Through Everyday Use

Although every function can be demonstrated during testing, the true value of the Smart Clock only becomes apparent after several weeks of everyday use.

Users gradually begin relying on features they previously thought were unnecessary.

Medicine reminders become routine.

Appointments are no longer forgotten.

The status LEDs become familiar.

The daily chimes quietly mark the passing of time.

Eventually, using the Smart Clock becomes second nature.

---

## 14.6 When Family Starts Asking for One

Perhaps the greatest compliment a builder can receive is when family members begin asking for their own Smart Clock.

This usually indicates that the system has moved beyond being an electronics project and has become something genuinely useful in everyday life.

Rather than simply building additional units for others, the author encourages family members and fellow hobbyists to use this manual as a guide to build their own.

In doing so, they gain not only a finished Smart Clock but also the satisfaction and experience that comes from constructing it themselves.

---

## Design Note – Built to Be Shared

The OUMA RIA SMART CLOCK was never intended to remain a one-of-a-kind project.

It is published with the hope that others will adapt it to suit their own homes, families and communities.

Some builders may add new sensors.

Others may redesign the enclosure.

Some may develop new software features.

This continued evolution is encouraged.

The Smart Clock is not a finished destination—it is a foundation for future ideas.

---

# Author's Reflection

One unexpected moment during development was when my wife no longer asked how the Smart Clock worked.

She simply used it.

Later, family members began asking for one of their own.

That was the moment I realised the project had crossed an invisible line.

It was no longer an electronics experiment.

It had become a useful part of everyday life.

---

# Chapter 15 – Commissioning, Testing & First Start-Up

## Introduction

Building the hardware is only half of the project.

The first power-up is the point where individual modules become a complete system. Taking the time to perform a careful, methodical first start-up greatly reduces the chance of damage and makes fault finding straightforward.

The Ouma Ria Smart Clock was designed so that every major subsystem can be tested individually before the complete firmware is loaded.

This chapter explains the sequence that was used during development and is recommended for every builder.

---

# Before Applying Power

Before connecting the power supply:

✔️ Check every solder joint.

✔️ Check that there are no solder bridges.

✔️ Verify the polarity of every electrolytic capacitor.

✔️ Verify all Dupont connectors.

✔️ Ensure every module shares a common ground.

✔️ Confirm the 5V and 3.3V rails are not accidentally connected together.

---

# Power Supply Test

Disconnect the ESP32.

Apply 12 V only.

Verify:

12 V present

↓

Buck Converter

↓

5.00 V

↓

ESP32 Expansion Board

↓

3.3 V

Only after both voltages are correct should the ESP32 be installed.

---

# ESP32 First Boot

Upload a simple Blink sketch first.

Confirm:

• Programming works

• USB communication works

• Serial Monitor operates correctly

Only then proceed to the full Smart Clock firmware.

---

# Test Each Module Individually

The development of this project proved one important lesson:

**Never debug everything at the same time.**

Instead, test every module independently.

Recommended order:

1. DS3231 RTC
2. BH1750 Light Sensor
3. LED Matrix
4. Status LEDs
5. Push Buttons
6. DFPlayer Mini
7. Web Server
8. Wi-Fi
9. Alarm Outputs
10. Pushover Notifications

If every module works individually, integrating them becomes much easier.

---

# Verify the Web Interface

After connecting to Wi-Fi:

Check:

✓ Current Time

✓ Temperature

✓ Uptime Counter

✓ NTP Status

✓ Wi-Fi IP Address

✓ Internet Status

✓ DFPlayer Status

These indicators provide immediate confirmation that the main subsystems are operating correctly.

---

# LED Status Bar Test

Verify every LED function:

Day indicator

Medicine reminders

Birthday reminders

Doctor countdown

Alarm status

Security status

24-hour mode

The Phone Mirror page should display exactly the same colours as the physical LED status bar.

---

# Automatic Brightness Test

Cover the BH1750 sensor.

The display should gradually dim.

Illuminate the sensor with a flashlight.

The display should smoothly brighten again.

The change should be progressive rather than abrupt, making the display comfortable to read under changing lighting conditions.

---

# Audio Test

Confirm that:

Westminster Chimes play.

Alarm sounds operate.

Reminder announcements play.

Button feedback sounds are heard.

If no sound is produced:

• Verify the SD card.

• Verify file numbering.

• Verify UART wiring.

• Confirm the DFPlayer reports **OK** on the web page.

---

# Internet Recovery Test

Disconnect Wi-Fi.

Observe:

Internet changes to NOT OK.

Reconnect Wi-Fi.

Internet returns to OK.

The Smart Clock automatically resumes normal operation.

---

# Loadshedding Test

This was an especially important test during development.

Simply remove mains power.

Restore power several hours later.

The Smart Clock should:

• reconnect to Wi-Fi

• synchronize with NTP

• restore the correct time

• continue operating without requiring user intervention.

Living in South Africa, where extended power interruptions are common, this behaviour became one of the design goals of the entire project.

---

# Final Functional Test

Before placing the Smart Clock into daily service, verify that:

✓ Time updates correctly.

✓ Temperature is displayed.

✓ Automatic brightness works.

✓ Medicine reminders operate.

✓ Status LEDs function.

✓ Audio playback works.

✓ Alarm activates correctly.

✓ Web interface is accessible.

✓ Calendar events display correctly.

✓ OTA updates are available.

✓ Pushover notifications are received.

Once these checks have been completed successfully, the Smart Clock is ready for everyday use.

---

## Author's Note

During the development of this project, the greatest amount of time was not spent writing software—it was spent finding poor-quality interconnecting leads, inexpensive clone modules, and small wiring faults.

Good-quality wiring, careful testing, and patience will save far more time than rushing to complete the build.

As the saying goes:

**"Build slowly. Test carefully. Enjoy the result for years."**

# PART II

# Builder's Guide

## Chapter 16 – Choosing Components

*"The quality of a project is determined long before the first wire is soldered."*

---

## 16.1 Introduction

The OUMA RIA SMART CLOCK was developed over an extended period using components obtained from a variety of suppliers.

While many modules performed exactly as expected, others introduced unnecessary complications due to inconsistent quality or undocumented differences between manufacturers.

The purpose of this chapter is to help builders make informed component choices based on practical experience rather than catalogue descriptions.

The recommendations that follow are based on the components used during the development of the original Smart Clock and the lessons learned along the way.

---

## 16.2 The ESP32

At the heart of the Smart Clock is an ESP32-WROOM development board.

The ESP32 was selected because it combines:

- Dual-core processing
- Built-in Wi-Fi
- Bluetooth capability
- Sufficient Flash memory
- Multiple GPIO pins
- Excellent support within the Arduino IDE
- A large and active development community

During development, the ESP32 proved exceptionally reliable and capable of managing all system functions simultaneously, including LED displays, Wi-Fi communication, audio playback, web services and reminder scheduling.

An ESP32 fitted to a breakout or expansion board is recommended, as it greatly simplifies wiring and future maintenance.

---

## 16.3 The RTC Module

Accurate timekeeping is essential for a reminder system.

Although the Smart Clock automatically synchronises with an Internet NTP server whenever Wi-Fi is available, the Real-Time Clock maintains accurate time during power interruptions and periods without Internet access.

A **DS3231** module is recommended because of its excellent long-term accuracy and integrated battery backup.

When mains power returns after an extended outage, the Smart Clock reads the RTC, reconnects to Wi-Fi, synchronises with the NTP server and resumes normal operation without requiring user intervention.

---

## 16.4 LED Displays

The display forms the visual centrepiece of the Smart Clock.

The original prototype uses WS2812B 8×8 LED matrices for the main display, together with additional WS2812B LEDs for the status indicators.

During development, one matrix failed following repeated power interruptions caused by load shedding.

Rather than waiting for a replacement, a standard WS2812B LED strip was cut into sections and soldered together in a serpentine arrangement to recreate the missing matrix.

This unexpected repair proved that individual LED strips can be adapted successfully when replacement matrices are unavailable.

It also demonstrated one of the guiding principles of the project:

*Solve the problem with the resources available.*

---

## 16.5 The BH1750 Light Sensor

The BH1750 digital light sensor was selected for automatic brightness control.

Unlike a simple LDR, the BH1750 provides stable, repeatable lux measurements over a wide range of lighting conditions.

The Smart Clock firmware uses these measurements to adjust the display brightness gradually, ensuring comfortable viewing during both daytime and night-time operation.

Separate day and night brightness settings are still provided.

These compensate for the different brightness characteristics of the various WS2812B modules used in the prototype, particularly the status LED strip, which produces noticeably more light than the display matrices.

---

## 16.6 The DFPlayer Mini

The DFPlayer Mini provides all spoken announcements, reminder tones and Westminster chimes.

This was one of the few components that required considerable investigation.

During development, an inexpensive clone module behaved differently from the original design.

The module would only recognise audio files when they were numbered sequentially, beginning with:

0001.mp3

0002.mp3

0003.mp3

and so on.

Although this behaviour differs from some genuine DFPlayer modules, the issue was successfully resolved through careful file organisation.

Builders are encouraged to purchase a genuine DFPlayer Mini where possible.

However, if only clone modules are available, they can still be used successfully provided their particular behaviour is understood.

---

## 16.7 Power Supplies

The Smart Clock operates from a regulated 12 VDC supply.

The author recommends the use of a fully enclosed desktop power adapter rated at approximately 10 A.

Compared with an open-frame switching power supply, an enclosed adapter offers:

- Improved electrical safety.

- Reduced risk of accidental contact with mains voltages.
- Simple plug-and-play installation.
- High reliability.

A high-current buck converter then provides the regulated 5 V supply required by the display and peripheral modules.

This arrangement has proven reliable during continuous operation.

---

## 16.8 Buy Once, Build Once

One of the most valuable lessons learned during development was that low-cost components can sometimes become the most expensive part of a project.

Poor-quality Dupont leads, inconsistent clone modules and unreliable connectors may save a small amount of money initially but often result in many hours of unnecessary fault-finding.

Where possible, builders are encouraged to purchase components from reputable suppliers and avoid the temptation to choose solely on price.

A reliable component is almost always less expensive than replacing several unreliable ones.

---

# Design Note – Experience Cannot Be Bought

Every recommendation in this chapter was earned through practical experience.

Some were learned after hours of troubleshooting.

Others only became apparent after weeks of everyday operation.

The purpose of sharing these experiences is not to criticise inexpensive components, but to help future builders recognise where investing in quality can significantly improve reliability and reduce frustration.

---

# Author's Reflection

When I first began building the Smart Clock, I believed that all modules advertised with the same name would perform identically.

Experience quickly proved otherwise.

Some inexpensive components worked perfectly.

Others behaved differently from the published documentation or required additional investigation before they could be used successfully.

Looking back, those challenges became valuable lessons.

If this chapter helps another builder avoid even one unnecessary week of fault-finding, then the experience has been well worth sharing.

# PART II – Builder's Guide

# Chapter 17 – Workshop Lessons Learned

*"Experience is the one component that cannot be purchased."*

---

## 17.1 Introduction

Every electronics project teaches valuable lessons.

Some are learned from successful experiments, while others come only after hours—or sometimes days—of troubleshooting.

The OUMA RIA SMART CLOCK was developed over several years of continuous improvement. During that time, many practical lessons were learned that are seldom found in datasheets or circuit diagrams.

This chapter shares those experiences with the hope of making the construction process easier for future builders.

---

## 17.2 Build One Section at a Time

One of the most effective development methods was to treat each subsystem as an individual project.

Rather than connecting every module at once, each section was assembled, tested and verified before moving to the next.

For example:

- Power supply

- ESP32
- RTC
- LED display
- BH1750 light sensor
- DFPlayer Mini
- Web interface
- Alarm outputs

By following this approach, faults can be isolated quickly and corrected before they affect the rest of the system.

---

## 17.3 Good Wiring Saves Days of Troubleshooting

One of the most time-consuming problems encountered during development was caused by inexpensive Dupont jumper leads.

Although they appeared perfectly normal externally, some were manufactured using only a single fine conductor.

Repeated movement caused intermittent internal breaks that were almost impossible to detect visually.

The solution was simple:

- Use quality Dupont connectors.
- Solder stranded instrument wire directly to the modules where practical.
- Secure cables so they cannot flex repeatedly.

This small improvement eliminated many intermittent faults.

---

## 17.4 Prototype First – Appearance Later

The original Smart Clock prototype was never intended to be a finished product.

Its purpose was to prove ideas, test software and evaluate hardware.

Blank PCB material, Veroboard and temporary wiring allowed components to be moved, replaced and modified easily throughout development.

Future builders should not be discouraged if their first prototype looks untidy.

A reliable prototype is far more valuable than an attractive one that cannot be serviced.

---

## 17.5 Don't Assume Every Module Is Identical

A lesson learned repeatedly was that two modules with the same product name may not behave identically.

This was particularly noticeable with inexpensive DFPlayer Mini modules.

Although sold under the same name, different versions required different approaches to file numbering and SD card preparation.

When troubleshooting unexpected behaviour, always consider the possibility that a module may differ from the one described in online examples.

---

## 17.6 Keep Notes

Throughout development, software settings, wiring changes and hardware modifications were carefully recorded.

These notes proved invaluable when revisiting a problem weeks or months later.

Builders are encouraged to keep a simple notebook containing:

- Pin assignments
- Wiring changes
- Calibration values
- Software revisions
- Test results

Good documentation is just as valuable as good soldering.

---

## 17.7 Test Under Real Conditions

Laboratory testing is only the beginning.

The Smart Clock was evaluated under everyday household conditions, including:

- Long periods of continuous operation.
- Extended load shedding.
- Automatic recovery after power restoration.
- Daily use by non-technical family members.

These practical tests revealed issues that would never have appeared during short bench tests.

---

## 17.8 Listen to Your Users

One of the most valuable sources of feedback came from the author's wife, Ria.

Many features that now form part of the Smart Clock were added because of everyday questions or practical needs that arose during normal use.

Examples include:

- The short-course antibiotic scheduler.
- Improvements to reminder displays.
- Clearer status indications.
- Simplified operation.

Real users often identify improvements that engineers overlook.

---

## 17.9 Don't Be Afraid to Improve

No engineering project is ever truly finished.

As new ideas emerge, the Smart Clock continues to evolve.

Future builders are encouraged to:

- Add new sensors.
- Improve the enclosure.
- Develop additional software features.
- Adapt the system to their own households and communities.

Every improvement contributes to the ongoing development of the project.

---

## 17.10 A Fault That Wasn't Where I Expected

Not every fault lives in the wiring.

One front-panel button began behaving strangely after the clock had been running for some time: pressing it produced no response, while at other moments the display would change on its own, as though the button had been pressed without anyone touching it.

The obvious suspects were checked first. The switch was replaced with a brand-new one. The wire running to it was removed completely. Even the ESP32 itself was swapped for a different board.

The fault remained.

Eventually, careful measurement traced the problem to the expansion board itself — a small amount of stray resistance on that one connection, almost certainly the result of the manufacturing process rather than anything in the design.

Rather than replace the board, the affected input was simply given a different, less critical job — one where an occasional false trigger would cause no harm — while a healthier pin took over the original function.

The lesson was a useful one: when a fault survives the replacement of every obvious component, it is worth questioning whether the fault is truly where it first appears to be. Sometimes the most efficient fix is not to chase the original cause, but to work around it sensibly.

# Design Note – Engineering Is a Journey

The OUMA RIA SMART CLOCK did not appear fully formed.

It evolved gradually, one improvement at a time.

Every challenge solved added another layer of reliability, usability or functionality.

The finished system is therefore not the result of a single design effort, but of many small refinements accumulated over time.

---

# Author's Reflection

Looking back, I realise that some of the greatest lessons came from mistakes.

A faulty jumper lead.

An unexpected DFPlayer clone.

A failed LED matrix during load shedding.

Each problem seemed frustrating at the time, but every solution made the Smart Clock better.

If there is one lesson I would pass on to future builders, it is this:

**Never become discouraged by a setback.**

Every problem solved is another step towards a more reliable design.

---

# PART II – Builder's Guide

# Chapter 18 – Troubleshooting

*"Every fault has a cause. Finding it is a process of elimination, not guesswork."*

---

## 18.1 Introduction

No electronics project is entirely free from problems during construction.

Fortunately, the vast majority of faults encountered while building the OUMA RIA SMART CLOCK are simple wiring errors, configuration issues or incorrect component installation.

This chapter provides a systematic approach to locating and correcting common problems experienced during both construction and operation.

One of the most important lessons learned during development was:

**Never change several things at once.**

Make one change.

Test it.

Observe the result.

Only then proceed to the next step.

This disciplined approach saves time and prevents unnecessary confusion.

---

# 18.2 The Smart Clock Will Not Power Up

### Symptoms

- Display remains dark.
- ESP32 Power LED is off.
- No Serial Monitor output.

### Possible Causes

- No 12 V supply.
- Blown fuse or polyfuse.
- Incorrect buck converter adjustment.
- Loose power connector.
- Reversed polarity.

### Checks

✔️ Measure 12 V input.

✔️ Measure 5 V output.

✔️ Measure 3.3 V at the ESP32 expansion board.

✔️ Verify all ground connections.

---

# 18.3 ESP32 Will Not Upload

### Symptoms

Arduino IDE reports upload errors.

### Checks

- Correct COM port selected.
- Correct ESP32 board selected.
- Good USB cable.
- Drivers installed.
- GPIO0 boot mode if required.

If necessary, press and hold the **BOOT** button while uploading.

---

# 18.4 RTC Time Is Incorrect

### Symptoms

Incorrect time after startup.

### Checks

- RTC battery fitted.
- SDA/SCL wiring.
- $I^2C$ address.
- NTP synchronisation.

Remember:

The RTC maintains time during power outages.

When Wi-Fi becomes available, the Smart Clock automatically synchronises with the NTP server.

---

# 18.5 BH1750 Brightness Does Not Change

### Symptoms

Display brightness remains constant.

### Checks

- I²C wiring.
- Correct sensor address (0x23 or 0x5C).
- Lux values visible in the Serial Monitor.
- Automatic brightness enabled.

Cover the sensor.

Then illuminate it with a flashlight.

Brightness should change smoothly.

---

# 18.6 DFPlayer Mini Produces No Sound

One of the more challenging issues encountered during development involved the DFPlayer Mini.

### Check:

✔️ UART wiring.

✔️ Speaker wiring.

✔️ SD card inserted.

✔️ FAT32 formatting.

✔️ Audio files copied sequentially.

✔️ Files named:

```
0001.mp3
```

```
...
```

`0027.mp3`

✔️ DFPlayer reports **OK** on the web page.

If using a clone DFPlayer Mini, remember that some versions require strict sequential numbering and physical file copy order on the SD card.

---

# 18.7 Westminster Chimes Play Incorrectly

### Symptoms

Wrong quarter-hour chime.

Incorrect hourly sequence.

### Checks

Verify:

- Correct MP3 numbering.
- Correct SD card copy order.
- Correct file contents.

The Smart Clock expects the standard 27-file audio package.

---

# 18.8 Wi-Fi Connects but Internet Is Not Available

### Symptoms

Wi-Fi OK.

Internet NOT OK.

### Explanation

A Wi-Fi connection does not necessarily mean Internet access.

The Smart Clock checks Internet connectivity independently before enabling services such as:

- NTP

- Pushover
- OTA updates

This distinction proved extremely useful during development and remains one of the system's status indicators.

---

# 18.9 Display Is Too Bright or Too Dim

### Possible Causes

- Incorrect BH1750 calibration.
- Day/Night brightness settings.
- Different LED batches.

Remember:

Different WS2812B modules may produce noticeably different brightness levels.

The adjustable Day and Night brightness settings allow the display and status LEDs to be balanced correctly.

---

# 18.10 Load Shedding Recovery

### Symptoms

Power restored after several hours.

### Expected Behaviour

The Smart Clock should:

- Restart automatically.
- Read the RTC.
- Connect to Wi-Fi.
- Verify Internet access.
- Synchronise with NTP.
- Resume normal operation.

No manual time adjustment should be required.

---

# 18.11 Random Problems

If behaviour appears inconsistent:

Do not immediately suspect the software.

Instead, check:

- Loose connectors.
- Poor Dupont leads.
- Bad solder joints.
- Power supply voltage.
- Common ground.
- SD card quality.
- Module quality.

Many apparently complex faults have surprisingly simple causes.

---

# Design Note – Troubleshoot Methodically

Successful troubleshooting is based on observation, measurement and patience.

Changing multiple settings at once often makes a fault more difficult to identify.

A careful, logical approach will almost always locate the problem more quickly.

---

# Author's Reflection

During the development of the OUMA RIA SMART CLOCK, there were times when a fault appeared to be caused by software, only to discover later that the real cause was a poor electrical connection or a low-quality component.

Those experiences reinforced an important lesson:

The simplest explanation is often the correct one.

Begin by checking the basics before assuming the problem is complex.

# PART II – Builder's Guide

# Chapter 19 – Maintenance, Upgrades & Future Development

*"A successful project is one that continues to improve."*

---

## 19.1 Introduction

Unlike many electronic devices that are considered complete once assembled, the OUMA RIA SMART CLOCK was designed to evolve.

Both the hardware and software were developed with future expansion in mind.

New reminder functions, improved software and additional sensors can be incorporated without redesigning the entire system.

This chapter describes routine maintenance and suggests possible future developments.

---

## 19.2 Routine Maintenance

Fortunately, the Smart Clock requires very little routine attention.

Periodic checks should include:

- Dust removal from ventilation openings.
- Inspection of wiring and connectors.
- Verification of power supply voltages.
- Checking the RTC backup battery.
- Confirming correct operation of the cooling airflow (if fitted).

A visual inspection every few months is usually sufficient.

---

## 19.3 SD Card Maintenance

The microSD card stores all audio announcements.

Although solid-state memory is generally reliable, it should be considered a replaceable component.

If unusual audio behaviour develops:

- Verify the SD card.
- Check file numbering.
- Confirm all 27 audio files are present.
- Replace the SD card if corruption is suspected.

Keeping a backup copy of the complete sound package on a computer is strongly recommended.

---

## 19.4 Firmware Updates

One advantage of the ESP32 platform is that new firmware can easily be installed.

As the project evolves, updates may include:

- Additional reminder functions.
- Improved display features.
- New web interface options.
- Additional alarm capabilities.
- Support for future sensors.

Builders are encouraged to keep notes describing any software changes they make.

---

## 19.5 Replacing Modules

The modular construction of the Smart Clock allows individual components to be replaced without rebuilding the entire system.

Modules such as:

- ESP32
- RTC
- BH1750
- DFPlayer Mini
- Power supply
- LED matrices

can normally be replaced independently.

Using connectors wherever practical greatly simplifies maintenance.

---

## 19.6 Battery Backup

Where a backup battery is installed, it should be inspected periodically.

Check:

- Terminal condition.
- Charging voltage.
- Battery capacity.
- Wiring integrity.

Reliable backup power is especially valuable in areas experiencing frequent power interruptions.

---

## 19.7 Future Hardware Expansion

The Smart Clock has been designed with expansion in mind.

Possible future additions include:

- Indoor air-quality monitoring.
- Carbon dioxide sensing.
- Blood oxygen monitor.
- Weather station.
- Rain gauge.
- Wind sensor.
- Additional motion detectors.
- Door and window monitoring.
- Bluetooth audio.
- Voice recognition.

The flexibility of the ESP32 platform makes these and many other additions possible.

---

## 19.8 Future Software Development

Software improvements may include:

- Additional reminder categories.
- More configurable schedules.
- Multiple user profiles.
- Voice prompts in different languages.
- Expanded web interface.
- Calendar synchronisation.
- Home automation integration.

Builders are encouraged to adapt the firmware to suit their own households and communities.

---

### 19.9 Open Development Philosophy

The author hopes that future builders will not only construct the Smart Clock but also improve it.

Constructive suggestions, new ideas and software enhancements are welcomed.

Every improvement strengthens the project and benefits the wider community of builders.

---

# Design Note – A Living Project

The OUMA RIA SMART CLOCK is not intended to remain static.

Every household is different.

Every family has different routines.

The project has therefore been designed as a flexible platform that can evolve over time while retaining the core philosophy of simplicity, reliability and practical usefulness.

---

# Author's Reflection

The greatest satisfaction does not come from completing a project.

It comes from seeing it become useful.

The Smart Clock began as a way to solve one practical problem.

Over time it became something much larger.

If future builders adapt the design, add new ideas or create features that better serve their own families, then the project will continue growing long after this book has been published.

That possibility is perhaps the most rewarding outcome of all.

---

# PART III – Reference Appendices

## Appendix A – Complete Bill of Materials (BOM)

*"A project is only as reliable as the components from which it is built."*

---

## A.1 Introduction

This appendix lists the principal components used in the construction of the **OUMA RIA SMART CLOCK**.

Equivalent components may be substituted where appropriate. However, the parts listed below have been successfully tested during the development of the original system and are therefore recommended.

---

# A.2 Main Electronics

| Qty | Component | Notes |
|---|---|---|
| 1 | ESP32 DevKit V1 (ESP32-WROOM-32) | Main controller |
| 1 | ESP32 Expansion Board | Simplifies wiring and provides regulated 3.3 V |
| 1 | DS3231 RTC Module | Battery-backed Real-Time Clock |
| 1 | BH1750 Light Sensor | Automatic brightness control |
| 1 | DFPlayer Mini | Audio playback module |
| 1 | MicroSD Card (Class 10, FAT32) | Stores the 27 MP3 files |
| 1 | MAX98357A *(optional)* | Future Bluetooth audio expansion |

# A.3 Display System

| Qty | Component | Notes |
|---|---|---|
| 4 | WS2812B 8×8 LED Matrix | Main display |
| 30 | WS2812B LEDs | Status indicators |
| 1 | Black translucent Perspex faceplate | LED diffuser and front panel |

# A.4 User Controls

| Qty | Component | Notes |
|---|---|---|
| As required | 12 × 12 mm tactile push-buttons | Chosen for positive tactile feedback |

# A.5 Audio & Alarm

| Qty | Component | Notes |
|---|---|---|
| 1 | IRLZ44N MOSFET | Drives external alarm |
| 2 | 10kΩ Resistor | One in series (GPIO to gate), one as gate pulldown to GND — the pulldown is required to prevent a false siren chirp at boot |
| 1 | 1N4007 Diode | Fitted across the siren/speaker terminals — protects the MOSFET from inductive kickback |
| 1 | 12 V Piezo Horn | High-power alarm output |
| 1 | Speaker | Connected to DFPlayer |

# A.6 Power Supply

| Qty | Component | Notes |
|---|---|---|
| 1 | 12 V 10 A Desktop Power Supply | Recommended |
| 1 | 8 A Buck Converter | 12 V to 5 V |
| 2 | 20 A Schottky Diodes | Automatic battery backup changeover |
| Optional | 12 V SLA or $LiFePO_4$ Battery | Backup supply |
| Several | Polyfuses | Circuit protection |

# A.7 Construction Materials

| Qty | Component | Notes |
|---|---|---|
| As required | Veroboard | Switches and small circuits |
| As required | Double-sided PCB | Wiring support and ground plane |
| As required | Instrument wire | Preferred over low-quality Dupont leads |
| As required | Heat-shrink tubing | Insulation |
| As required | Cable ties | Cable management |
| As required | Standoffs and spacers | Mounting boards |

# A.8 Enclosure

Recommended:

- 3D printed enclosure
- Black translucent Perspex front panel
- Steel base plate
- Rubber feet

- Rear connectors for:
    - Power
    - Alarm output
    - USB programming

---

# A.9 Optional Accessories

- PIR Sensor
- External temperature sensor
- Bluetooth speaker
- Wi-Fi antenna
- Oxygen monitor (future project)
- Additional alarm indicators

---

# Purchasing Advice

During the development of the Smart Clock, it became clear that spending slightly more on quality components often saved many hours of troubleshooting.

Where possible:

- Purchase from reputable suppliers.
- Avoid extremely low-cost clone modules unless you are prepared to test them thoroughly.
- Use good-quality connecting leads and connectors.
- Label components as they are installed.

The small additional cost is usually recovered many times over through improved reliability and reduced fault-finding.

# PART III – Reference Appendices

## Appendix B – ESP32 GPIO & Pin Assignment Reference

*"A clear wiring plan is worth more than a hundred photographs."*

---

### B.1 Introduction

This appendix documents the GPIO assignments used in the reference design of the OUMA RIA SMART CLOCK.

During development, every effort was made to group related functions together and reserve spare GPIO pins for future expansion.

Builders may modify these assignments if required, but should ensure that any software changes remain consistent with the hardware wiring.

---

## B.2 Main Processor

| GPIO | Function | Notes |
|---|---|---|
| GPIO16 | DFPlayer RX | ESP32 receives data from DFPlayer |
| GPIO17 | DFPlayer TX | ESP32 transmits commands to DFPlayer |
| GPIO21 | I²C SDA | RTC & BH1750 |
| GPIO22 | I²C SCL | RTC & BH1750 |

*(Continue with the remaining GPIO assignments from your final circuit.)*

---

## B.3 I²C Devices

| Device | Address |
|---|---|
| DS3231 RTC | 0x68 |
| BH1750 | 0x23 *(default)* |

Both devices share the same I²C bus.

---

## B.4 Serial Interfaces

| Interface | Purpose |
|---|---|
| USB Serial | Programming and debugging |
| UART2 | DFPlayer Mini |

Using a dedicated UART for the DFPlayer avoids conflicts with USB programming and simplifies software development.

---

## B.5 Power Rails

| Voltage | Purpose |
|---|---|
| 12 V | Main power input, siren, backup battery |
| 5 V | LED displays, DFPlayer, peripheral modules |

| Voltage | Purpose |
|---|---|
| 3.3 V | ESP32 logic, RTC, BH1750 and other low-voltage sensors |

The ESP32 expansion board provides convenient distribution of the regulated 3.3 V supply to compatible modules.

---

# B.6 LED Connections

Document:

- Main display data line.
- Status LED data line.
- 470 Ω series resistors.
- Filter capacitors.
- Power injection points.

A simplified wiring diagram is recommended for builders assembling the display.

---

# B.7 Push Buttons

Document every button:

| Button | Function |
|---|---|
| Menu | Opens configuration menu |
| Up | Increase value |
| Down | Decrease value |
| Enter | Confirm selection |
| Alarm | Alarm control |
| ... | ... |

Using descriptive names rather than only GPIO numbers makes both the software and wiring easier to understand.

---

# B.8 Expansion Board

One feature that distinguishes the reference design is the use of an ESP32 expansion board.

The expansion board offers several advantages:

- Simplifies wiring.
- Provides convenient power distribution.

- Reduces the number of soldered connections to the ESP32 itself.
- Makes maintenance easier.
- Allows modules to be replaced without disturbing the main controller.

Although the ESP32 DevKit already contains its own 3.3 V regulator, the expansion board provides an organised and reliable method of distributing both 5 V and 3.3 V to the connected modules.

---

## B.9 Grounding

A reliable ground system is essential.

Recommendations:

- Use a common ground point.
- Keep high-current LED returns separate from sensitive signal wiring where practical.
- Ensure every module shares a common ground reference.
- Check continuity before applying power.

Many intermittent problems can be traced to poor grounding rather than faulty software.

---

## Design Note – Label Everything

One lesson learned during the construction of the prototype was the value of clear labelling.

As additional modules were added over time, identifying connectors quickly became increasingly important.

Simple labels on cables and connectors greatly simplify future maintenance and fault-finding.

# PART III – Reference Appendices

# Appendix C – Wiring & Connector Reference

*"Good wiring is invisible when it works and invaluable when it doesn't."*

---

# C.1 Introduction

The reliability of any electronic system depends not only on the circuit design but also on the quality of its wiring.

Throughout the development of the OUMA RIA SMART CLOCK, careful attention was given to cable routing, connector selection and power distribution.

Many intermittent faults were eventually traced to poor-quality wiring rather than faulty electronic components.

The recommendations in this appendix are based on practical experience gained during the construction of the original prototype.

---

# C.2 Wiring Philosophy

Every wire should have a purpose.

Avoid unnecessary cable lengths.

Avoid loops.

Keep wiring neat.

If someone else opens the enclosure five years from now, they should immediately understand how it is wired.

A neat layout is not simply cosmetic—it makes maintenance, troubleshooting and future upgrades much easier.

---

# C.3 Power Wiring

Three voltage rails are used:

| Voltage | Purpose |
|---|---|
| 12 V | Main supply, alarm output and battery backup |
| 5 V | LED displays, DFPlayer Mini and peripherals |
| 3.3 V | ESP32 logic, RTC and BH1750 |

Power wiring should always use conductors of adequate size.

Voltage drop becomes increasingly important where LED currents are involved.

---

# C.4 Signal Wiring

Low-current signal wiring should be kept separate from higher-current LED and alarm wiring wherever practical.

Examples include:

- I²C (RTC and BH1750)
- UART (DFPlayer Mini)
- Button inputs
- Sensor signals

Keeping these cables tidy helps reduce electrical noise and makes servicing easier.

---

# C.5 Connector Selection

During development, one of the most valuable lessons concerned connector quality.

Although inexpensive Dupont jumper leads appeared satisfactory, some contained only a single fine conductor.

Repeated movement caused intermittent faults that were extremely difficult to locate.

For this reason, the following approach is recommended:

- Use good-quality Dupont connectors where removable connections are required.
- Where practical, solder stranded instrument wire directly between the module and the connector.
- Avoid repeated flexing of cables.

Good wiring saves countless hours of fault-finding.

---

# C.6 Expansion Board

The ESP32 expansion board provides an organised connection point for the entire system.

Advantages include:

- Clearly labelled connections.
- Convenient 3.3 V distribution.
- Easier maintenance.
- Reduced mechanical stress on the ESP32.
- Simplified replacement of modules.

Using an expansion board is strongly recommended for builders constructing the Smart Clock.

---

# C.7 Cable Routing

Try to route wiring in logical groups.

For example:

Power wiring

↓

Display wiring

↓

Sensor wiring

↓

Audio wiring

↓

Control wiring

Keeping related cables together improves both appearance and maintainability.

Cable ties or reusable cable straps may be used to keep wiring secure.

---

# C.8 Grounding

Every module must share a common ground reference.

The prototype makes extensive use of a common ground plane formed from double-sided PCB material.

This provides:

- Low impedance.
- Good mechanical support.
- Convenient solder points.
- Reduced voltage drop.

Good grounding contributes significantly to reliable operation.

---

# C.9 Decoupling Capacitors

Small decoupling capacitors should be fitted close to modules where recommended.

The LED display power rails should also include suitable electrolytic capacitors to absorb current surges during rapid LED changes.

These simple components improve system stability.

---

# C.10 Label Everything

As the project expanded, additional modules, sensors and connectors were added.

Simple labels on cables and connectors greatly reduced maintenance time.

A small investment in labelling during construction saves considerable time later.

---

# C.11 Keep Spare Connectors

It is worth keeping a small supply of:

- Dupont housings
- Crimp pins
- JST connectors
- Heat-shrink tubing
- Cable ties

These inexpensive items are often the first to be needed during modifications or repairs.

---

# Design Note – Reliability Begins with Wiring

Many builders naturally focus on the ESP32, sensors and software.

In reality, a reliable wiring system is just as important.

Most long-term problems arise from poor electrical connections rather than software defects.

Good wiring should therefore be regarded as part of the design rather than simply a method of joining components together.

---

# Author's Reflection

During the construction of the prototype, I spent far more time tracing wiring faults than writing software.

Looking back, most of those problems could have been avoided by using better-quality connectors and taking a little more time during assembly.

Today, if I were to build another Smart Clock, I would make exactly the same circuit—but I would finish the wiring even more carefully.

Good workmanship is one of the simplest ways to improve reliability.

# PART III – Reference Appendices

# Appendix E – Software Configuration & Customisation

*"The Smart Clock is designed to adapt to its owner—not the other way around."*

---

## E.1 Introduction

One of the key design goals of the OUMA RIA SMART CLOCK was flexibility.

Although the system performs many functions immediately after installation, nearly every operating parameter can be customised to suit the needs of an individual household.

Most configuration changes can be made without modifying the source code.

---

## E.2 Wi-Fi Configuration

Before the Smart Clock can access Internet services, the following information must be configured:

- Wi-Fi Network Name (SSID)
- Wi-Fi Password

Once connected, the Smart Clock automatically:

- Synchronises its clock with an Internet NTP server.
- Enables Pushover notifications.
- Confirms Internet connectivity.
- Continues normal operation after future power interruptions.

---

## E.3 Time and Date

The user may configure:

- 12-hour or 24-hour clock display.
- Time zone.
- Daylight Saving Time (where applicable).
- Date format.

During normal operation, the RTC maintains accurate time whenever Internet access is unavailable.

---

## E.4 Display Settings

The display may be adjusted to suit personal preference.

Available settings include:

- Day brightness.
- Night brightness.
- Automatic brightness adjustment using the BH1750 light sensor.
- Temperature display interval.
- Display timeout (if enabled).

Separate day and night brightness settings allow different LED batches to be visually balanced, particularly where replacement WS2812B strips have been used.

---

## E.5 Medicine Scheduler

The medicine scheduler is one of the Smart Clock's principal functions.

The current software supports:

- Multiple daily medication reminders for ongoing (chronic) medication.
- **Two independent short-course medication schedules** for temporary treatments such as antibiotics.

Each reminder can be individually configured with:

- Start date.
- End date (for short courses).
- Reminder times.
- Reminder description.
- Audible notification.

The addition of a second short-course scheduler reflects a real-world requirement identified during everyday use.

---

## E.6 Calendar and Appointments

Users may configure reminders for:

- Birthdays.
- Doctor's appointments.
- Personal reminders.
- Household events.

Each entry includes:

- Date.
- Time.
- Description.
- Reminder notification.

---

## E.7 Timers

The timer system supports a variety of household activities.

Examples include:

- Kitchen timers.
- Braai timers.
- Smoker timers.
- General countdown timers.

Because the timer names and durations are user-defined, the system can easily be adapted to other hobbies or activities.

---

## E.8 Audio Settings

Users may customise:

- Alarm volume.
- Reminder sounds.
- Westminster chimes.
- Hourly chimes.
- Voice announcements.

Audio files may be replaced with custom recordings provided the required filename sequence is maintained.

---

## E.9 Pushover Notifications

When Internet access is available, the Smart Clock can transmit notifications for:

- Medicine reminders.
- Smoker timer reminders.
- General alarms.
- Other configurable events.

This allows important reminders to be received even when the user is away from the Smart Clock.

---

## E.10 Future Configuration Options

The software has been designed with future expansion in mind.

Additional configuration pages may be added as new features are developed without altering the overall structure of the system.

---

# Design Note – Configuration Without Complexity

Although the Smart Clock provides many configurable options, care has been taken to keep the user interface simple and intuitive.

The aim has always been to make advanced functionality available without overwhelming users who simply want reliable reminders and clear information.

---

# Author's Reflection

One of the guiding principles during development was that the Smart Clock should adapt to the household in which it is installed.

No two families follow exactly the same routines.

By making the software highly configurable, the system can serve the needs of older people living independently, busy families, hobbyists and many others without changing the underlying hardware.

# Builder's Guide

---

# Chapter 20 – Final Testing & Commissioning

This is the chapter where the builder proves that everything works **before fitting the front panel and closing the enclosure**.

After the Smart Clock has been fully assembled, every subsystem should be tested individually before the enclosure is closed. Taking the time to perform these checks can save many hours of fault-finding later.

## Step 1 – Power System

Connect the 12 V power supply.

Confirm:

- Buck converter adjusted to **5.0 V**
- ESP32 powers up normally
- Expansion board power LED illuminates
- No component becomes hot

---

## Step 2 – ESP32 Startup

Open the Serial Monitor.

Confirm that:

- ESP32 boots without reset loops
- Wi-Fi connects
- IP address is displayed
- NTP time synchronises
- Internet status reports **OK**
- DFPlayer reports **OK**

---

# Step 3 – LED Matrix

Verify:

- Clock time displays correctly.
- Colon flashes normally.
- Brightness changes automatically with room lighting.
- Day and night brightness settings operate correctly.

---

# Step 4 – Status LEDs

Check every status indication:

- Current weekday pulses softly.
- Medicine LEDs illuminate correctly.
- Alarm status LEDs respond to arm/disarm.
- Countdown indicators operate correctly.

---

# Step 5 – Push Buttons

Operate every button.

Confirm that each performs its assigned function without false triggering.

If any button behaves erratically, first inspect the wiring and verify that the required external **10 kΩ pull-up resistors** are fitted where specified.

---

# Step 6 – PIR Sensors

Walk through each protected area individually.

Confirm that:

- Correct zone activates.
- Alarm responds correctly.
- False triggers do not occur.

---

## Step 7 – Audio System

Verify:

- DFPlayer initialises successfully.
- All required MP3 files play.
- Westminster chimes operate.
- Alarm sounds correctly.
- Voice prompts are clear.

---

## Step 8 – Siren Output

Test the emergency siren only in a safe location.

Verify:

- IRLZ44N MOSFET switches correctly.
- Piezo horn operates at full volume.
- PWM output is functioning correctly.

**Warning:** The 150 W piezo horn is extremely loud. Avoid prolonged exposure at close range.

---

## Step 9 – Load Shedding Test

Disconnect the mains supply.

Confirm:

- Backup battery supplies the system.
- Clock continues running.
- No reboot occurs.

Reconnect mains power.

Confirm:

- Battery charging resumes.
- Time remains correct.
- NTP automatically resynchronises.

---

# Step 10 – Final Functional Test

Finally, test the Smart Clock exactly as it will be used.

Create:

- A medicine reminder.
- A short-course reminder.
- A birthday reminder.
- A doctor appointment.
- A calendar event.
- An alarm.
- A security alarm test.

The Smart Clock should now operate exactly as intended.

---

## A Note from the Designer

The Smart Clock was not developed only on a workbench. It has been used daily in a real home environment and refined over many months of practical use. Several software features—including the dual short-course medicine scheduler—were added only after real-life situations highlighted their value. Builders are encouraged to use the system for a few weeks and adapt it to suit their own household's needs.